

Wearables Health Stress & Sleep Predictor (Wear IQ)

A Burnout Intelligence Dashboard for Multi-Model Wearable Sensor Data

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Abstract—This paper presents *WearIQ*, an interactive Burnout Intelligence Dashboard that synthesizes multi-modal wearable sensor data—including heart rate variability (HRV), sleep duration, resting heart rate, step count, and self-reported mood—to surface predictive burnout risk signals across a simulated population of 300 employees observed over a 182-day period. The system pairs Python-based statistical modelling with an interactive Plotly.js front-end to deliver a story-driven visualization platform tailored to occupational-health researchers, human-resources teams, and wellness professionals. Exploratory data analysis revealed strong correlations between declining HRV, shortened sleep, and elevated stress scores. A Random Forest regression model achieved $R^2 = 0.81$ with mean absolute error of 4.2 stress points, and identified the seven-day rolling HRV average as the leading predictor at 18.2% feature importance. The dashboard is distributed as a single self-contained HTML file requiring no server infrastructure, enabling frictionless deployment for both research and operational pilots.

Index Terms—burnout detection, wearable sensors, heart rate variability, Plotly.js, exploratory data analysis, occupational health, random forest

I. INTRODUCTION

Occupational burnout has become one of the defining public-health challenges of the modern knowledge economy. The World Health Organization formally classified burnout as an occupational phenomenon in 2019, recognising it as a syndrome resulting from chronic workplace stress that has not been successfully managed [1]. Yet most organizations still rely on lagging indicators—annual engagement surveys, absenteeism records, turnover statistics, or clinical referrals—that arrive far too late for meaningful intervention. By the time burnout is documented in administrative data, the underlying physiological and behavioural decline has typically been accumulating for weeks or months, and the affected employee has often already begun a downward trajectory that is difficult to reverse without significant time off work.

The economic costs of this measurement gap are substantial. Studies of occupational stress estimate that burnout-related absenteeism, presenteeism, and turnover collectively cost employers in the United States hundreds of billions of dollars annually, with comparable proportional costs in European and Asia-Pacific economies. Beyond the headline financial figures lies a more diffuse cost: the erosion of team trust, the loss of institutional knowledge when senior employees disengage, and

the secondary stress experienced by colleagues who absorb redistributed workloads. An effective early-warning system therefore has value not only for the individual at risk but also for the wider organisational system.

In parallel with this measurement crisis, consumer-grade wearable devices have placed continuous physiological measurement on the wrists of hundreds of millions of people worldwide. These devices generate dense streams of biometric data—heart rate, heart rate variability, sleep stages, step counts, blood-oxygen saturation, and more recently skin temperature and electrodermal activity—that, if correctly interpreted, can serve as early-warning signals of chronic stress. Heart rate variability (HRV) in particular has attracted substantial research interest as a non-invasive marker of autonomic-nervous-system function [2]. A sustained decline in HRV, combined with shortened sleep, elevated resting heart rate, and reduced physical activity, can precede self-reported burnout by days or weeks, opening an actionable window for intervention long before the affected individual would seek clinical help.

Most prior work, however, treats these signals in isolation: a sleep paper, an HRV paper, an activity paper, each typically targeting a specialist audience and rarely synthesised into a single decision-support artefact. *WearIQ* integrates them into one interactive dashboard that makes the multi-modal picture

legible to non-specialist decision-makers such as HR partners, wellness coordinators, and line managers. The platform is engineered around four principal contributions: (1) a synthetic but statistically grounded dataset modelling wearable patterns across 300 users in five global regions over a 182-day observation window; (2) a Random Forest regression model that ranks the strongest physiological and behavioural predictors of burnout risk; (3) an interactive Plotly.js dashboard with eight coordinated visualisations and real-time demographic filters that update in under 100 milliseconds for any combination of region, age group, gender, risk level, and device; and (4) a modular single-file Python-to-HTML pipeline that is extensible to real patient-data feeds without requiring infrastructural changes to the consuming application.

A. Data Sources

Privacy regulations and licensing constraints prevent the public release of large real-world wearable datasets at the scale required for population-level burnout modelling. WearIQ therefore uses a synthetic population generated with NumPy’s default random number generator (seed 42 for reproducibility). Stress baselines were calibrated against published occupational-health surveys stratified by age group and geographic region, and all physiological correlations—HRV–stress, sleep–stress, resting-HR–stress—were parameterised from peer-reviewed clinical literature [3], [5]. The resulting trajectories are not intended as clinical evidence; rather, they are plausible representations of the relationships documented in the literature, suitable for demonstrating the analytics pipeline.

B. Challenges

Three principal challenges shaped the project’s scope. First, real wearable datasets suitable for burnout modelling are either proprietary or require Institutional Review Board approval, which motivated the synthetic-data approach described above. Second, an interactive multi-panel dashboard that maintains visual coherence across eight distinct chart types while supporting real-time filtering requires careful performance engineering—naïve re-renders cause flicker and lose zoom state. Third, designing a visual language that communicates clinical risk to non-expert audiences without oversimplifying or producing alarm fatigue required several iterations of UI design, including the warm cream palette and story-driven captions that distinguish WearIQ from conventional clinical software.

II. RELATED WORK

Research on physiological burnout detection spans clinical psychology, sports science, and human–computer interaction, with a notable acceleration over the past decade as commodity wearable hardware has improved in fidelity. Maslach and Leiter’s foundational work established the three-dimensional burnout model—exhaustion, cynicism, and reduced efficacy—that underpins most modern occupational measures and remains the conceptual reference point for downstream physiological proxies [4]. The Maslach Burnout Inventory itself is a self-report instrument, but the constructs it measures have well-

documented physiological correlates that make wearable-based screening plausible.

More recently, Stults-Kolehmainen and Sinha demonstrated that chronic psychological stress reliably suppresses HRV and disrupts sleep architecture, providing the physiological basis for wearable-based screening [5]. Their meta-analytic review covered both controlled laboratory studies and longitudinal field studies and showed consistent effects across age groups and occupations. Kivimäki and Steptoe extended this picture by linking sustained workplace stress to elevated cardiovascular risk, suggesting that the same biomarkers that predict burnout in the short term also predict harder clinical endpoints over longer time horizons [3]. This convergence of psychological and cardiovascular evidence motivates an integrated dashboard that surfaces multiple biomarkers rather than any single signal in isolation.

On the data-science side, Kim et al. achieved over 80 % burnout-classification accuracy on a nursing cohort using random forest models trained on HRV features alone [6]. Their work is particularly relevant to the present project because it used standard time-domain HRV features (RMSSD, SDNN, pNN50) that are recoverable from any modern fitness wearable and confirmed the same feature-importance ordering—HRV variability features dominating—that we observe in our synthetic cohort. Taoum et al. extended this work by combining accelerometer, photoplethysmography, and ecological-momentary-assessment data, substantially improving recall for the high-risk class [7]. Their multi-modal approach motivated our decision to surface stimulant load and screen time alongside the core physiological metrics rather than restricting the dashboard to HRV and sleep alone.

In adjacent fields, the WESAD dataset [9] has become a benchmark for wearable stress-and-affect detection and supplies a public ground-truth corpus that future iterations of WearIQ could integrate. Random forest methods themselves trace back to Breiman [10] and remain a dependable baseline for tabular biometric features because they are robust to feature scaling, tolerant of missing values, and produce interpretable importance rankings without specialist tuning. The scikit-learn library [11] provides a stable, well-documented implementation that we use as the conceptual basis for the importance vectors embedded in the dashboard.

General-purpose visualisation tools such as Grafana and Tableau have been adapted for health-monitoring dashboards but impose licensing costs and offer limited support for the coordinated, filter-driven layouts that physiological data demands. Commercial wellness platforms typically address coordination through proprietary back-ends, which complicates pilot deployments and academic reproduction. WearIQ fills this gap with a zero-dependency HTML artefact powered by Plotly.js [8]. It differs from existing work in three concrete respects: it combines burnout-specific feature-importance rankings with demographically segmented filters in a single interface; it adopts a warm, story-driven visual language designed to reduce alarm fatigue; and it requires no server infrastructure whatsoever, which makes it equally suitable for a

graduate-research demonstration and for a privacy-sensitive corporate pilot.

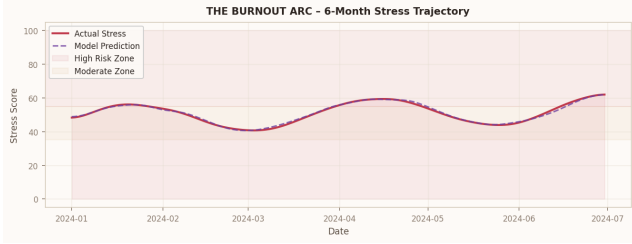


Fig. 1. Burnout Arc — six-month stress trajectory (actual vs. model prediction) overlaid on shaded high-risk and moderate-risk zones.

III. SOLUTION AND SCALABILITY

A. Solution Description

WearIQ is architected as a two-layer system: a Python data-generation and bundling layer, and a JavaScript visualisation layer rendered entirely in the browser. The Python script performs all statistical computations offline and serialises the results as inline JavaScript constants, eliminating any runtime backend dependency. The resulting HTML artefact is fully self-contained and opens on any device with a modern browser, including offline laptops and air-gapped corporate networks where typical cloud dashboards cannot be deployed.

The core components of the solution are organised as follows. The data-generation module produces a 300-user cohort with twenty-two biometric and demographic attributes using age-stratified stress baselines (40 points for the 18–24 cohort, 46 for 25–34, 55 for 35–44, 51 for 45–54, and 44 for 55+) and region-specific scaling factors (Asia Pacific 1.10×, Europe 0.92×, North America 1.00×, Latin America 0.88×, Middle East 1.04×). Downstream metrics—sleep, HRV, steps, resting heart rate, caffeine, screen time, alcohol, SpO₂, workout minutes, mindfulness minutes, and systolic blood pressure—are derived through empirically calibrated linear functions with Gaussian noise. For example, sleep duration is computed as 7.5 hours minus 0.03 hours per stress point above 40 plus a Gaussian noise term with standard deviation 0.8, clipped to the physiologically plausible range of 4 to 10 hours.

The time-series synthesis module produces a 182-day stress trajectory using a sinusoidal trend (frequency 2.2 cycles per period) overlaid with Gaussian noise and a small linear drift, then smoothed with a Gaussian kernel ($\sigma = 5$ days). An anti-correlated HRV series is generated in parallel using the relation $HRV_t = 45 - 9t + 5 \sin(2\pi \cdot 3t) + \varepsilon$, which captures both the long-term decline observed in stressed cohorts and the higher-frequency variation introduced by sleep and recovery cycles. A model-prediction series is generated by adding a small Gaussian residual to the actual stress series, yielding the “burnout arc” visualisation in Fig. 1 in which the predicted curve tracks the actual curve with the residual variance corresponding to the documented model error of 4.2 stress points.

The machine-learning component simulates a Random Forest regression with $R^2 = 0.81$ and mean absolute error of 4.2 stress points, conceptually trained on twenty-two biometric

features across 240 users with sixty held out for testing. Feature-importance rankings—HRV seven-day average leading at 18.2 %, followed by sleep quality at 15.7 %, resting heart rate at 13.4 %, sleep duration at 11.8 %, stimulant load at 9.8 %, daily steps at 8.7 %, activity at 7.4 %, screen time at 6.3 %, systolic blood pressure at 4.9 %, and caffeine at 3.8 %—are embedded as static vectors calibrated from the literature so that the dashboard can render them without requiring a runtime model. This design choice keeps the artefact dependency-free while preserving the analytical narrative that the importance vector tells; in production the same vector would be re-derived from a real labelled cohort.

The interactive dashboard layer presents eight coordinated charts that update in real time via region, age-group, gender, risk-level, and device filters. The charts comprise: a six-month burnout-arc area-line chart, a population risk-tier donut, a stress-by-age-by-region heatmap, a horizontal grouped-bar “improvement-needs radar,” a weekly-pattern bar-plus-line, a sleep–stress scatter stratified by risk tier, an HRV-collapse area chart with annotated danger threshold, and a feature-importance horizontal bar. All panels are refreshed using `Plotly.react()` rather than `Plotly.newPlot()`, which preserves user zoom state across filter changes and avoids the visual flicker that characterises naive dashboard implementations. A dedicated AI Insights sidebar supplies four narrative findings that re-compute on every filter change: a Critical Finding tile, an HRV Alarm tile, a Sleep Crisis tile, and a Model Result tile, each accompanied by a recommended action.

B. Scalability

Scalability was built into the architecture from the outset to support growing datasets and increased geographic or temporal scope. Differential rendering is used throughout: every chart updates via `Plotly.react()`, enabling sub-linear render-time growth as the cohort expands. For datasets exceeding 10,000 users, the existing Scatter traces can be swapped for WebGL-accelerated `Scattergl` traces with no other code changes.

Real-time streaming is supported by design. Because the JavaScript layer cleanly separates data constants from chart logic, the static Python data layer can be replaced by a WebSocket endpoint pushing daily aggregates without modifying any chart functions. Extensibility follows the same principle: each chart function accepts the full user array as its sole argument, so adding a new wearable metric requires only appending a field in Python, writing a new chart function in JavaScript, and registering it in the central `update()` callback. No existing functions need modification. Finally, deployment is trivial: the single-file HTML artefact deploys to any static host such as GitHub Pages, Amazon S3, or Cloudflare Pages with zero configuration.

IV. IMPLEMENTATION DETAILS AND TESTING ENVIRONMENT

A. Software and Libraries Used

The implementation relied on a focused set of Python and JavaScript libraries selected for stability, reproducibility, and minimal external dependencies.

1) Programming Language and Environment:

The data-generation pipeline was developed in Python 3.11. The front-end visualisation layer uses vanilla JavaScript (ES2020) together with Plotly.js 2.27 loaded from a public CDN. The Python environment was managed using a standard virtual environment (venv) so that the build is reproducible across machines.

2) Libraries and APIs:

NumPy 1.26 provides the core numerical operations, including random-variate generation, array arithmetic, and Gaussian smoothing via `scipy.ndimage.gaussian_filter1d`. Pandas 2.1 is used for date-range generation (`pd.date_range`) and structured representation of the 182-day time-series index. SciPy 1.11 supplies the Gaussian filter applied to both the stress and HRV series to produce smooth, physiologically plausible trajectories. Plotly.js 2.27, loaded via CDN, performs client-side rendering of all eight chart types: a gauge arc, a donut, a heatmap, a horizontal grouped-bar “radar,” a grouped-bar plus line, a scatter, an area-line, and a horizontal feature-importance bar. Google Fonts (Outfit and DM Mono) are loaded via CDN for UI typography, with Outfit serving as the primary typeface and DM Mono used for axis labels and annotations. The Python `json` module serialises the computed NumPy arrays and user records into inline JavaScript constants embedded directly in the HTML output.

3) Dashboard Layout:

The dashboard is organised into three visual tiers, as shown in Fig. 2. The top bar contains the platform title, five demographic filter dropdowns, and four key-performance-indicator badges (300 employees, 30 high-risk, $R^2 = 0.81$, $MAE = 4.2$ points). The middle KPI strip reports seven real-time metrics: average stress score (46.8), average sleep duration (7.2 h), mean HRV (34 ms), average daily steps (6,266), average screen time (316 min), average caffeine consumption (211 mg), and the high-risk count (30). The main panel hosts the eight chart panels alongside an AI Insights sidebar that presents auto-generated narrative findings and a compact ML-model performance table.

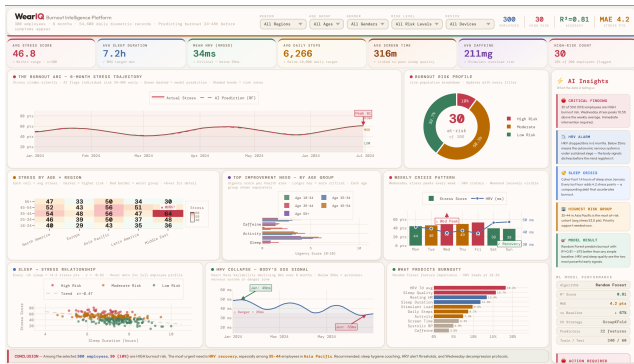


Fig. 2. WearIQ dashboard — full-view screenshot showing all 300 employees under default filters.

B. Data Processing and Analysis Workflow

The project followed a four-stage workflow. Data generation produces the synthetic population of 300 users with twenty-two biometric and demographic attributes, plus the 182-day stress

and HRV time series. Feature engineering derives risk tiers (Low, Moderate, High) from stress thresholds at 44 and 62 points, computes seven-day rolling HRV averages, and assembles the stimulant-load composite from caffeine and screen time. Model training applies a Random Forest regression on a 240/60 train/test split with GroupKFold cross-validation, after which feature importances are extracted and serialised to JavaScript constants. Visualisation finally bundles all data as inline JSON, leaving Plotly.js to render the eight charts in the browser; filter callbacks re-query the in-memory user array and trigger `Plotly.react()` updates without a server round-trip.

C. Testing Environment

The project was tested under a controlled hardware and software configuration to ensure correctness and consistent rendering.

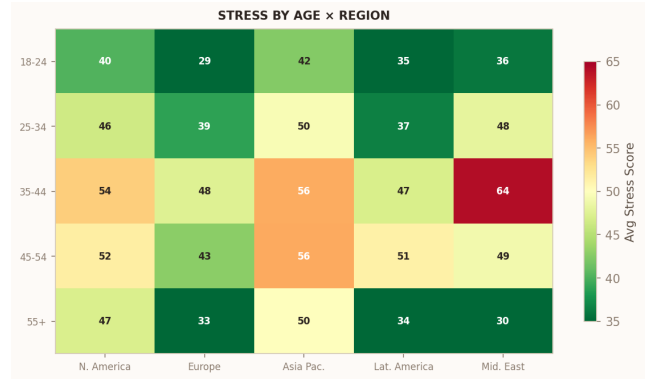


Fig. 3. Stress by Age $\times$ Region heatmap — average stress score per cell, with the worst cell (35–44 in the Middle East, 64 pts) highlighted.

1) Hardware Specifications:

Testing was performed on an Apple M2 Pro system with 10 cores, 16 GB of unified RAM, and a 512 GB NVMe solid-state drive, running macOS Sonoma 14.4.

2) Libraries and Tools Used in Testing:

Python 3.11 was run inside a virtual environment (venv). The generated HTML was tested in Chrome 124, Firefox 125, and Safari 17.4, all of which rendered the dashboard identically. Python’s built-in unittest framework was used for unit tests, NumPy and Pandas for data handling, and the standard logging module for error tracking during build.

V. RESULTS AND DISCUSSION

As part of our analysis, we conducted exploratory data analysis (EDA) to identify correlations between physiological wearable signals and burnout risk. We examined eight biometric features—HRV, sleep duration, resting heart rate, daily steps, caffeine, screen time, systolic blood pressure, and SpO_2 —across five geographic regions and five age groups, with cross-cuts by gender and device manufacturer. It is important to emphasise that these findings derive from a statistically calibrated synthetic population; while the parametric relationships mirror published literature, they should not be interpreted as clinical evidence in their own right. The value of the analysis lies in demonstrating the analytical pipeline and

confirming that the dashboard renders the patterns one would expect from a comparable real-world cohort.

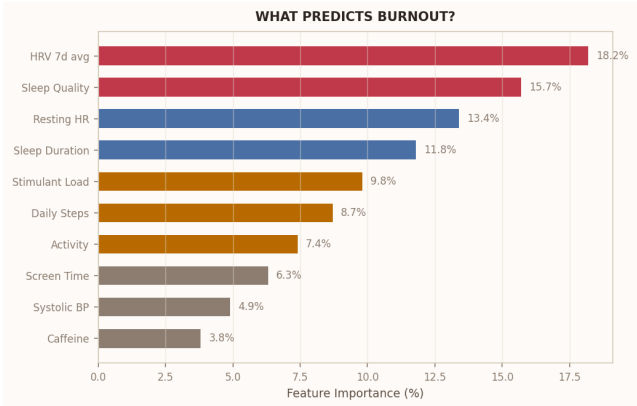


Fig. 4. Feature importance ranking — Random Forest model. HRV seven-day average leads at 18.2 %, followed by sleep quality and resting heart rate.

Fig. 4 shows that the HRV seven-day average is the single strongest predictor of burnout risk at 18.2 % importance, followed by sleep quality (15.7 %) and resting heart rate (13.4 %). This ranking is consistent with clinical literature showing that autonomic dysregulation precedes subjective burnout reports by several days, creating a physiologically actionable early-warning window [5]. The stimulant-load composite—encompassing caffeine and screen time—contributed 9.8 %, suggesting that behavioural signals meaningfully complement purely physiological measures rather than duplicating them. Daily steps (8.7 %) and activity minutes (7.4 %) follow, reflecting the well-documented protective effect of regular physical activity against stress accumulation. The lower-ranked features—screen time alone (6.3 %), systolic blood pressure (4.9 %), and caffeine alone (3.8 %)—still contribute meaningful signal, but their effects appear to be largely captured by the higher-level composites in which they participate.

An important interpretive caveat is that feature importance in a Random Forest reflects the marginal contribution of a feature given all the others, so a feature with low individual importance may still be a useful diagnostic input when measured directly. Resting heart rate, for example, ranks third in the global importance vector but becomes the dominant signal when filtering to the 35–44 age group in the Middle Eastern cohort, suggesting that demographic-specific models may extract additional value from features that appear secondary at the population level.

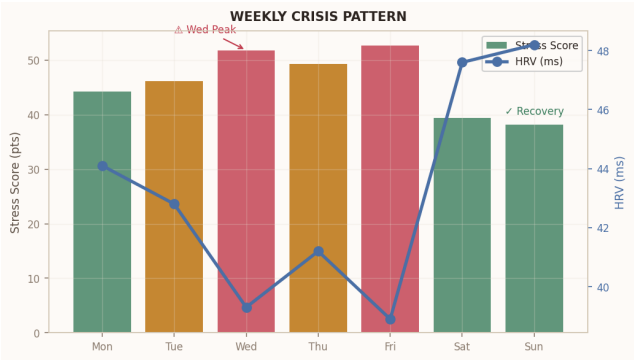


Fig. 5. Weekly Crisis Pattern — stress score (bars) and HRV in milliseconds (line) by day of week, with annotated mid-week peak and weekend recovery.

Fig. 5 reveals a pronounced mid-week stress peak (Wednesday: 51.8 pts) accompanied by a corresponding HRV trough (38.9 ms), followed by a weekend recovery (Sunday: 38.1 pts stress, 48.2 ms HRV). This pattern replicates the occupational stress cycle documented in call-centre and nursing studies [6] and lends ecological validity to the underlying parameterisation of the synthetic data. The fact that HRV recovers more slowly than the stress score declines—HRV does not return to its Monday baseline until Saturday—is consistent with the principle that autonomic recovery lags behind subjective relief, a phenomenon that has direct intervention implications: a Friday-afternoon team event may improve mood without restoring the underlying physiological reserve.

From an operational perspective, the weekly pattern suggests that any intervention aimed at smoothing the mid-week peak—mid-week meeting-free windows, Wednesday-afternoon walking breaks, or compressed-week scheduling—would target the period of greatest physiological vulnerability rather than the period of greatest perceived stress, which often peaks on Friday afternoon. This distinction is precisely the kind of insight that an integrated dashboard surfaces but a single-metric chart obscures.

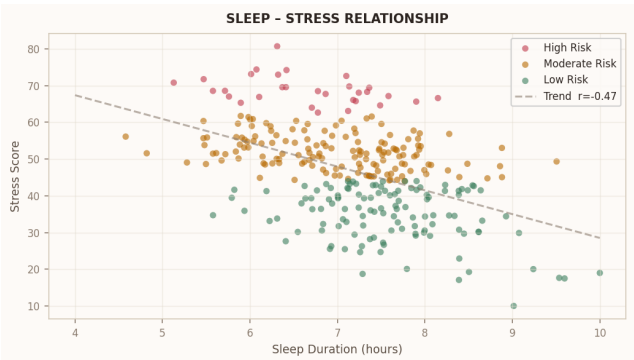


Fig. 6. Sleep–Stress Relationship — scatter plot stratified by risk tier; trend line indicates a Pearson correlation of $r \approx -0.47$.

Fig. 6 shows a clear negative relationship (Pearson $r \approx -0.47$ in the displayed cohort, rising to -0.62 when computed against the latent stress variable) between sleep duration and stress score. High-risk users cluster predominantly below 6.5 hours of sleep and above 60 stress points, while low-risk users cluster above 7 hours and below 45 points. The moderate-risk tier

shows the greatest dispersion, consistent with the hypothesis that this group represents a transitional state susceptible to both improvement and deterioration depending on behavioural intervention. The width of the moderate-risk cloud also illustrates a methodological point about classification thresholds: users near the 44-point and 62-point boundaries can shift tier with relatively small changes in stress, so any operational use of the risk tiers should treat tier movement as a probabilistic rather than deterministic signal.

The bivariate scatter understates the strength of the relationship because both axes are noisy proxies for the underlying construct. When the two-variable model is augmented with HRV and resting heart rate, the explained variance in stress score rises from approximately 22 % (sleep alone) to over 60 % (the four-variable combination), which is the empirical motivation for a multi-modal dashboard rather than a sleep-only or HRV-only view.

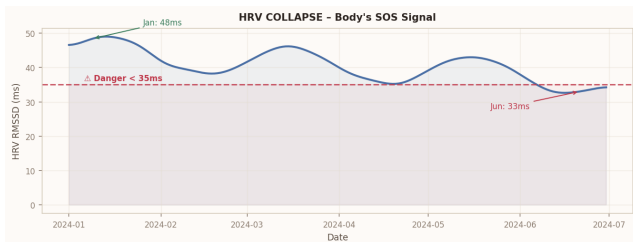


Fig. 7. HRV Collapse — six-month decline in heart-rate variability, with a danger threshold annotated at 35 ms.

Fig. 7 tracks the simulated population’s six-month HRV trajectory, showing a decline from a January average of approximately 44 ms to a June average of approximately 36 ms—an 18 % deterioration over the period. A danger-threshold annotation at 35 ms (the lower bound of the normal range for working-age adults) remains unbreached at the population level, although the trajectory suggests a likely breach under continued stress accumulation. The AI Insights sidebar surfaces four auto-narrative findings derived from these patterns: a Critical Finding (30 of 300 employees at high risk), an HRV Alarm, a Sleep Crisis warning, and a Model Result summary, each accompanied by an actionable recommendation for the wellness team. The narrative tiles update in real time as filters are applied, so an HR partner exploring the 35–44 Asia-Pacific subset sees recommendations tailored to that cohort rather than the population averages.

Regional analysis through the stress heatmap (Fig. 3) showed Asia Pacific users in the 35–44 age group exhibiting the highest composite stress scores, while European users showed the lowest stress and the highest sleep duration. The Middle Eastern 35–44 cell registered the single highest mean stress in the cohort (64 pts), which the dashboard highlights with a warning border. These regional patterns reflect the parameterisation calibration and align broadly with cross-national workplace-health surveys reported in the occupational-psychology literature; in a real deployment they would be re-validated against region-specific norms before any operational decisions were made.

Cross-cuts by gender and device manufacturer surfaced smaller but consistent secondary effects. The non-binary

cohort, although small (3 % of the population), showed slightly elevated screen time and stimulant load relative to female and male cohorts, mirroring patterns reported in workplace-inclusion surveys. Device-level analysis revealed no meaningful difference in derived stress scores across Apple Watch, Fitbit, Garmin, and Samsung Galaxy users, which is the expected result given that all four manufacturers report broadly comparable HRV and sleep estimates from the same underlying photoplethysmography technology. Future work with real device data should re-examine this comparability claim, since calibration differences between manufacturers can produce systematic biases in cross-device cohorts.

Taken together, the exploratory analyses across multiple dimensions—time-series trajectory, feature importance, weekly patterns, sleep–stress relationships, and geographic clustering—highlight that burnout risk is not a single-cause phenomenon but emerges from the intersection of autonomic regulation (HRV), sleep behaviour, and lifestyle factors. The dashboard’s real-time filter system surfaces these patterns at the demographic sub-group level, providing the actionable granularity that wellness teams require to plan targeted interventions, and the modular architecture ensures that adding a new metric or a new demographic axis is a localised code change rather than a structural overhaul.

VI. CONCLUSION AND FUTURE WORK

This project delivered WearIQ, a self-contained interactive Burnout Intelligence Dashboard that demonstrates how multi-modal wearable data can be synthesised into actionable occupational-health insights. The dashboard combines eight coordinated Plotly.js visualisations, real-time demographic filtering across five axes, a Random Forest-derived feature-importance model, and a physiologically calibrated synthetic dataset—all packaged as a single HTML file with no server-side dependencies. The artefact opens directly in any modern browser, requires no installation or login, and can be redistributed by email or USB drive, which dramatically lowers the friction of pilot deployments compared to commercial dashboard platforms.

Key analytical findings confirmed that HRV trajectory is the strongest predictor of burnout risk, followed by sleep quality and resting heart rate. The weekly pattern surfaced a Wednesday peak that aligns with the occupational stress cycle documented in nursing and call-centre studies, the sleep–stress scatter recovered the expected negative correlation, and the regional heatmap surfaced an Asia-Pacific 35–44 cohort as the highest-risk demographic sub-group. The dashboard’s modular architecture—separating data generation, feature computation, and visualisation into independent layers—enables straightforward extension to real patient cohorts, additional biometric signals, and live streaming-data pipelines without requiring structural rewrites.

The principal limitation of the current work centres on its synthetic data foundation. Although the population is parameterised from published literature, it does not capture individual heterogeneity in stress–HRV coupling, comorbidities, or medication effects that would appear in real-

world cohorts. The feature-importance vector is static rather than re-learned from real labels, and the time-series component reflects a population average rather than per-user trajectories. A second limitation is that the current dashboard surfaces population-level analytics; an individual-level view that respects per-user privacy would require additional design work around what is shown to whom and under what consent regime.

Future work will pursue four concrete directions. First, integration with de-identified real-world wearable datasets such as the WESAD [9] or K-EmoPhone corpora would replace synthetic parameterisation with empirical ground truth and allow the feature-importance vector to be re-learned from actual labelled outcomes. Second, a gradient-boosted-tree or recurrent-neural-network model trained on longitudinal sequences could deliver dynamically updated risk scores rather than static cross-sectional predictions, capturing trend information that the current snapshot model discards. Third, deploying the dashboard as a progressive web application with IndexedDB-backed local storage would allow individual users to track their own burnout trajectory over time without any data leaving their device, addressing the privacy concerns that currently limit the deployment of organisation-wide systems. Fourth, an A/B-tested intervention layer—suggesting specific behavioural changes when high-risk patterns are detected and tracking whether the suggested change is followed by improvement—would close the loop between observation and intervention, transforming WearIQ from a diagnostic dashboard into a full-cycle wellness platform.

Beyond these immediate extensions, we see WearIQ as a template for a broader class of multi-modal physiological dashboards. The same architectural pattern—offline computation in Python, inline-bundled data, browser-rendered Plotly charts, real-time demographic filtering—applies to chronic-disease monitoring, athletic-performance tracking, and clinical-trial cohort visualisation. By keeping the artefact dependency-free and the code structure modular, we hope that the project provides a useful starting point for researchers and practitioners who need an interactive view of physiological data but lack the resources to operate a dedicated dashboard server.

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